



v1.0 · Whitepaper · Don't Generate. Buy.

SoundHub

A token-powered, DAW-native marketplace for **verified, finished sounds**.

Versioned assets, escrowed purchases, automatic creator splits and DAO governance — with a storefront that lives *inside* the DAW.

Network: Base (EVM, L2) **Token:** SND · ERC-20, 1,000,000 fixed **Status:** Private beta · testnet live

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1. Abstract

SoundHub is a marketplace where producers buy ready-made, human-made, **production-ready sounds** — presets, sample packs, patches and stems — paid for with the SND token on **Base**. What makes the purchase trustworthy is not reputation alone but **verification**: a DAW-parsing engine reads the project file and proves what is inside the asset before the buyer pays. What makes it fast is **placement**: the storefront is a native panel inside Ableton Live, at the moment of intent.

The system is organised in three layers:

LAYER	WHAT IT DOES	TRUST MECHANISM
Studio (off-chain)	DAW parsing & verification, smart diffs, repos, versioning	Content-addressed storage (SHA-256), structured metadata
Tokenized (on-chain)	SND payments, escrow marketplace, Release NFTs, DAO	Smart-contract escrow, ERC-2981 royalties, Governor + timelock
Distribution (in-DAW)	SoundHub panel inside Ableton Live: catalog, BPM-aware suggestions, buy & load	Same DAW engine — verify, recommend, version

This document specifies the architecture, the economic model, the verification pipeline and the governance of the platform, and honestly scopes what is live today versus what is proposed.

2. The problem: generation is cheap, time is not

Music production's real bottleneck is **time, not tools**. Two forces dominate:

- **Generation is cheap; finished sound is expensive.** AI and endless preset-tweaking produce *a* sound — rarely *the* sound. Producers burn hours per sound and still end up with inconsistent, unverifiable results.
- **Buying finished sound is untrusted.** Marketplaces are fragmented, licenses are unclear, contents are unverifiable, provenance is missing, and creators get

paid late or never.

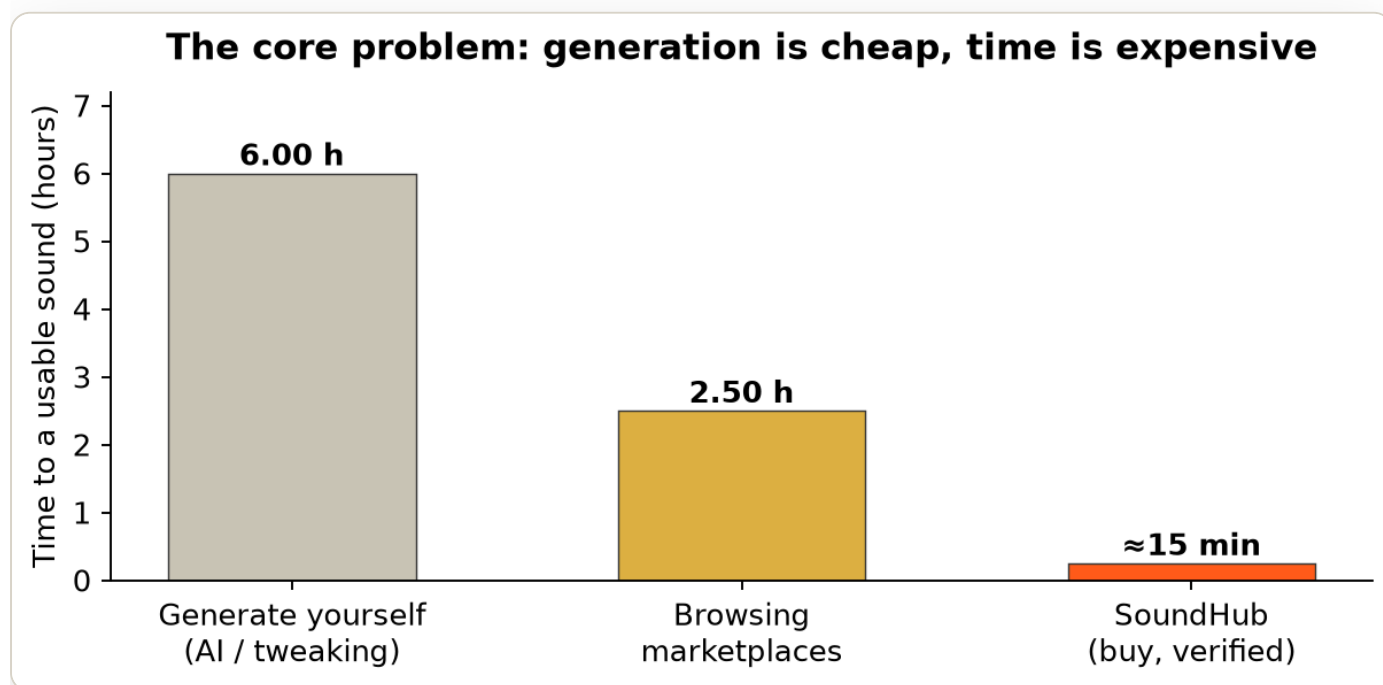


Figure 1 — The core problem: hours spent generating vs. seconds spent buying a verified sound.

There is no attribution infrastructure: independent producers and sound designers have no lightweight way to prove authorship, set licenses, or get paid for reuse. Code got repositories, attribution and licensing decades ago; sound is still sold like loose files in random chats.

3. The solution: three layers, one product

SoundHub flips the equation — **don't generate, buy**. Finished, human-made sounds become a token-powered marketplace on Base. The core loop:

Create → List → Verify → Buy → Get paid.

1. A sound designer creates a finished preset, kit, or pack (in Ableton, FL Studio, Cubase, REAPER).
2. SoundHub parses and verifies it — the DAW engine reads BPM, tracks, plugins, samples.
3. The asset is listed for SND with a license tier and version history.
4. A producer buys — SND goes into escrow; the buyer gets the asset + license.

5. Payment settles — buyer confirms, or the dispute window passes, and the creator is paid automatically.

The three layers share one engine: the DAW parsers that *verify* assets for sale also power the in-DAW *recommendations* and the *smart diffs* of project versions.

Verification, recommendation and versioning are three jobs of the same parsing pipeline.

4. Studio layer — versioning & the DAW engine

4.1 Content-addressed versioning

Every commit stores a **full snapshot** of the project tree. Files are stored **content-addressed**: the storage key is the SHA-256 hash of the file bytes.

$$\text{blob_key} = \text{SHA-256}(\text{file bytes})$$

(1)

Because the key is a deterministic function of the content, identical files deduplicate automatically — a snapshot in which little changed costs almost nothing. A commit is then an immutable mapping from paths to blob keys:

$$\text{commit}_n = \{ \text{path} \mapsto \text{SHA-256}(\text{bytes}(\text{path})) \mid \forall \text{path} \in \text{tree}_n \}$$

(2)

This gives history a simple, correct model with cheap snapshots; branches are named pointers, and the move to a DAG (with merges) is the natural next step. Every decision on the platform is additionally recorded in a **SHA-256 decision ledger**.

4.2 The DAW engine — verification before purchase

The DAW engine is what makes buying finished sound trustworthy. SoundHub parses the four major project formats instead of treating them as binary, so a

buyer sees exactly what is inside an asset *before* paying:

FORMAT	DAW	PARSED DATA
.als	Ableton Live	BPM, time signature, tracks, devices, plugins, samples
.cpr	Cubase	tempo, tracks, VST3/VST plugins, referenced samples
.rpp	REAPER	tempo, signature, tracks, FX chain (PARAM lines)
.flp	FL Studio	version, project name, author, tempo, channel count

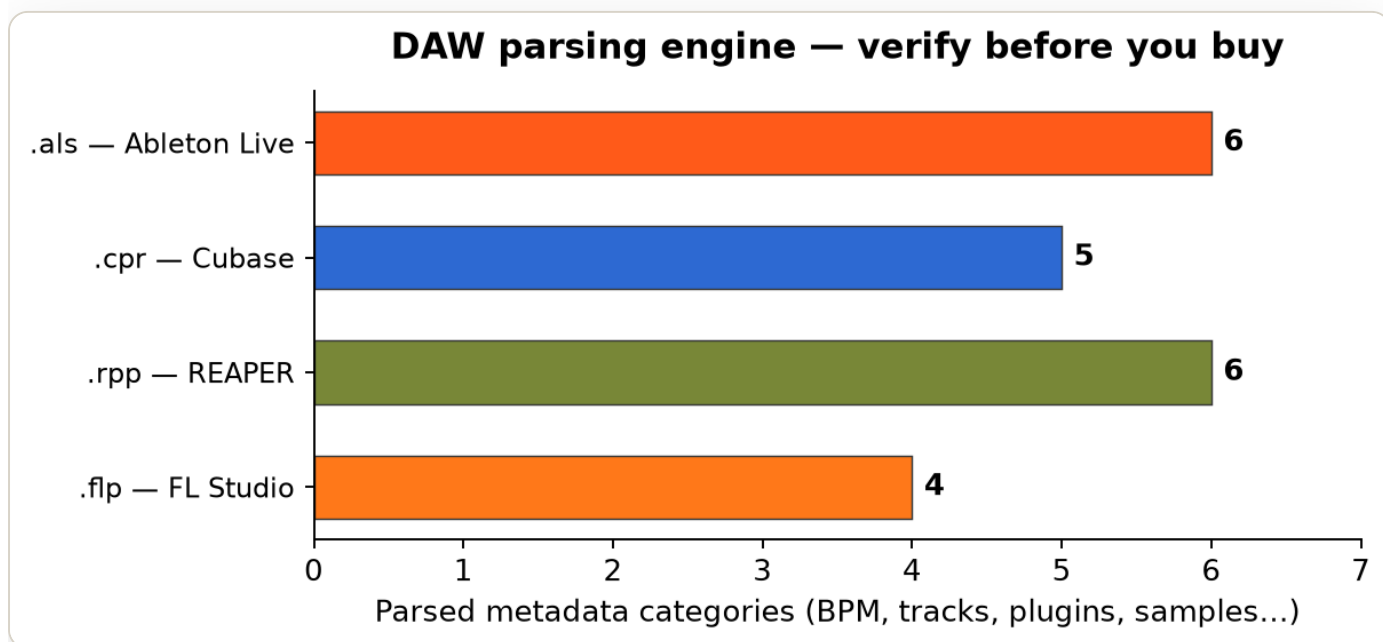


Figure 2 — Parsed metadata categories per DAW format.

Each parser extracts a `DAWInfo` structure — BPM, time signature, tracks (with kind and devices), plugins (with settings where the format exposes them) and referenced samples. The same normalised structure is used for asset verification, for smart diffs and for recommendations.

4.3 A worked example — a real Cubase session

The screenshot below is a real Cubase 13 session pushed to SoundHub via `snd push`: 126 BPM, five tracks, eight plugins (Serum, Vital, FabFilter, SSL, Waves...), four referenced samples — all parsed from the `.cpr` XML.

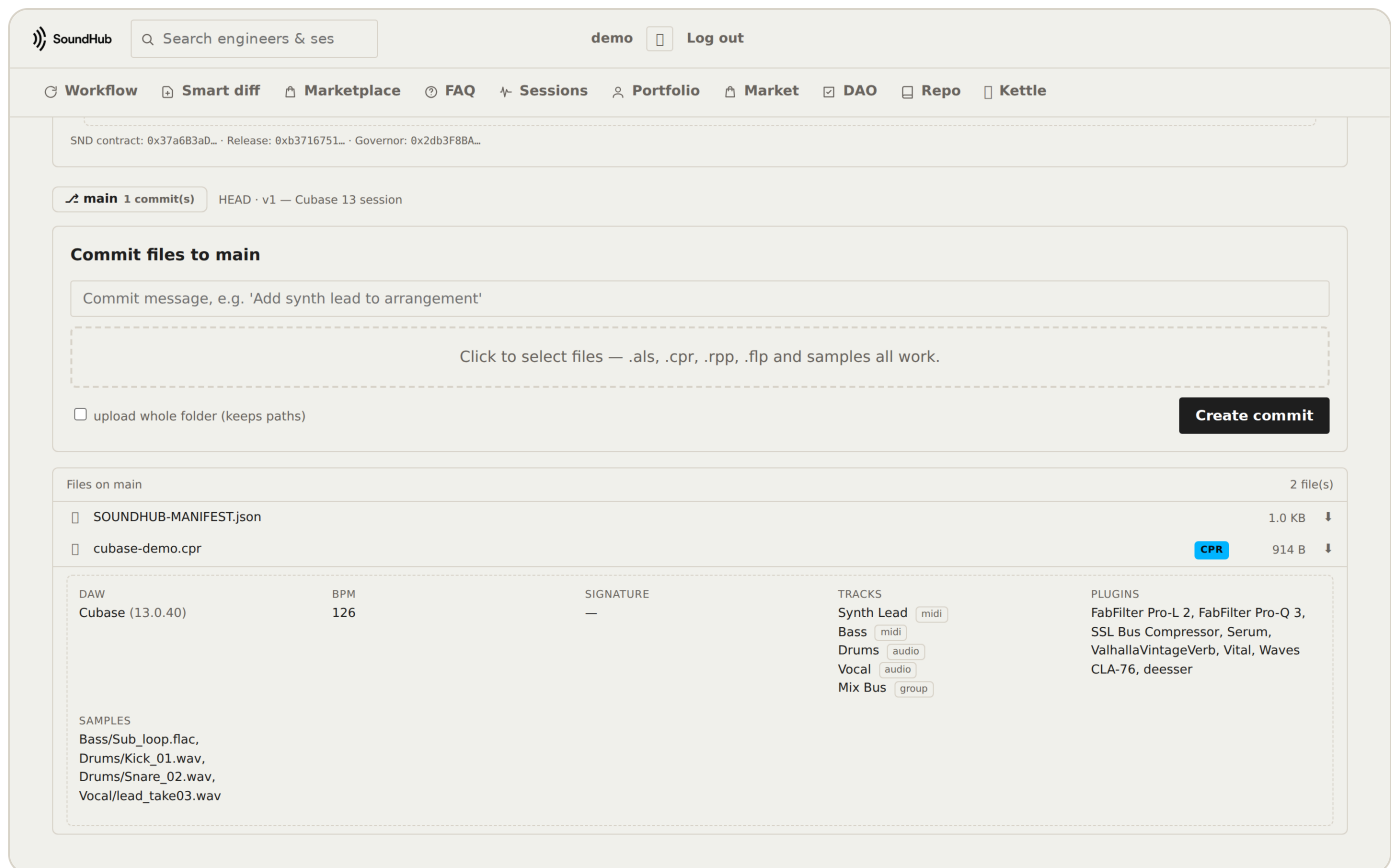


Figure 3 — DAW-parsed metadata for a Cubase 13 project in the SoundHub repo UI.

5. Smart diffs — structured, not "binary changed"

Where GitHub says *"binary file changed"*, SoundHub says:

```
BPM 128 → 132
+ track Pad (midi)
+ track Vocal Chops (audio)
+ plugin Vital
+ sample VocalChop_01.wav
```

The diff engine produces two views:

- **Summary diff** — structured metadata comparison: tempo, time signature, DAW version, added/removed tracks, plugins and samples.
- **Raw unified diff** — over the normalised content (pretty-printed XML, text, or hex for binary formats such as `.flp`).

Formally, for two versions *A* and *B*, the summary diff is the symmetric difference of their parsed feature sets:

$$\Delta(A, B) = (\text{features}(A) \setminus \text{features}(B)) \cup (\text{features}(B) \setminus \text{features}(A)) \quad (3)$$

where **features**($\cdot$) includes the BPM, signature, track set, plugin set and sample set extracted by the DAW engine. Scalar features like BPM are reported as **old** $\rightarrow$ **new** ; set features as **+/-** entries. The same engine, two jobs: **verify assets for sale** and **version projects in collaboration**.

The screenshot shows the SoundHub web interface. At the top, there's a search bar and navigation links like Workflow, Smart diff, Marketplace, FAQ, Sessions, Portfolio, Market, DAO, Repo, and Kettle. The main content area displays a project named 'demo / Aurora Night' with a 'Public' status. Below the project name, there's a 'pushed via snd' message and a 'projects' link. A 'Delete repo' button is visible in the top right. The interface is divided into sections: 'On-chain release' and 'Commit files to main'. The 'On-chain release' section includes a form to turn the project into an on-chain release NFT, with fields for the project name (Aurora Night), a value (500), and a collaborators list. A 'Mint release NFT' button is at the bottom right of this section. The 'Commit files to main' section has a text area for a commit message and a file selection area. A 'Create commit' button is at the bottom right of this section. The interface also shows a 'main' branch with 2 commit(s) and a 'HEAD' pointing to 'Round 3 candidate'.

Figure 4 — A DAW project as a versioned repo: file tree with parsed tracks and plugins.

6. Tokenized layer — SND, escrow, NFTs

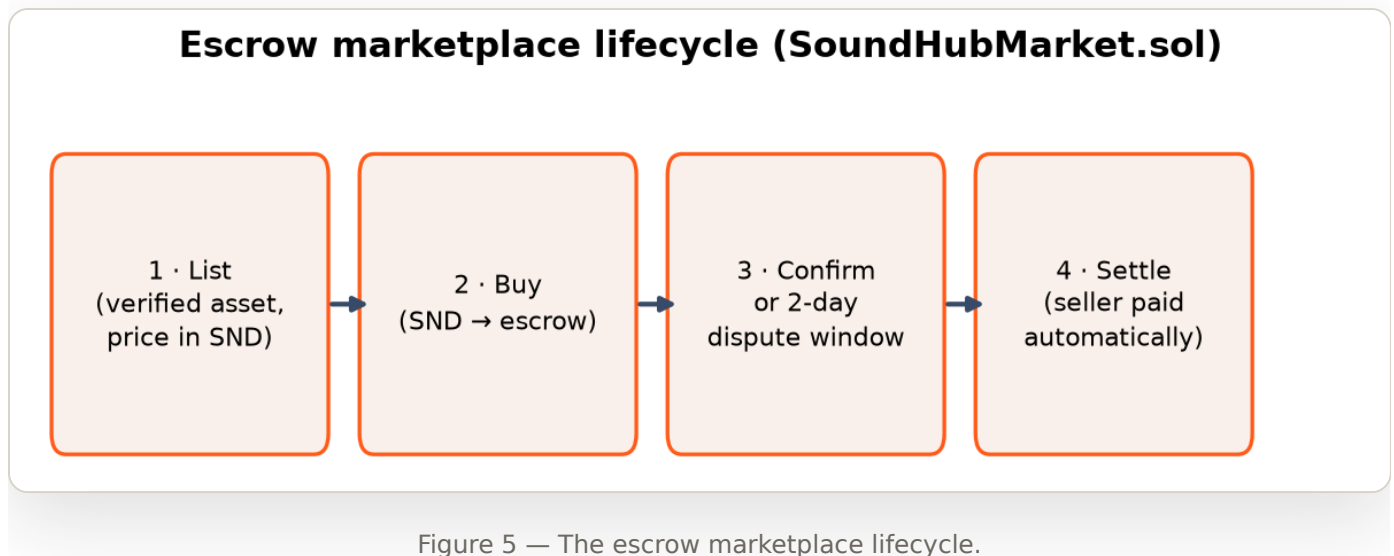
6.1 Wallet identity

Users sign in with any EVM wallet via an EIP-191 personal-sign challenge verified server-side. No password is needed; the wallet address becomes the account. The server verifies that the recovered signer matches the claimed address:

$$\text{signer} = \text{ecrecover}(\text{hash}(\text{nonce}), \sigma) \wedge \text{signer} = \text{address} \quad (4)$$

6.2 The escrow marketplace — SoundHubMarket.sol

Finished sounds are bought with SND, not generated. The lifecycle:



1. **List** — a seller publishes a verified asset with a price in SND, a license tier (personal / commercial / sync / exclusive) and a pointer to the file.
2. **Buy** — the buyer's SND goes into escrow; the buyer receives the asset and license.
3. **Settle** — the buyer confirms receipt (instant payout to the seller) or the **2-day dispute window** passes (seller withdraws).
4. **Refund** — inside the window a buyer can request a refund; the arbiter (owner today, DAO tomorrow) resolves it.

No trust-me selling. No chargebacks. Payment is automatic and on-chain.

6.3 SND — the platform token (SND.sol)

- ERC-20 with **permit** (gasless approvals) and **ERC20Votes** (on-chain governance voting).
- **Fixed supply: 1,000,000 SND**, minted once at deployment. No mint function.
- **Primary role: the payment rail** — every marketplace purchase is denominated in SND. Secondary roles: license fees, tips, token-gated premium

features, staking, and DAO voting power via delegation.

6.4 Release NFTs — SoundHubRelease.sol

A finished project can be minted into a **Release NFT** (ERC-721 + ERC-2981):

- **Royalty** — configurable per release (default 5%), paid to the minter on secondary sales via ERC-2981.
- **Collaborator split** — revenue split with basis-point weights summing to 10,000, recorded on-chain at mint. E.g. producer 70% / vocalist 30%.
- **Treasury** — fans fund a release directly with **ETH or SND** (`fund / fundWithSND`).
- **Order-independent claiming** — collaborators claim their proportional share of the treasury at any time; claiming early does not short-change later claimers (per-collaborator accounted claims).

Formally, for a release with collaborator weights w_i (summing to 10,000), the claimable amount for collaborator i from a treasury T is:

$$\text{claim}_i = T \cdot \frac{w_i}{\sum_j w_j} - \text{already_claimed}_i \quad (5)$$

Because each claim records only the collaborator's own withdrawal, a later claim is unaffected by earlier ones — the payout is order-independent.

7. Tokenomics — fixed supply, governed allocation

SND has a **fixed supply of 1,000,000** — and its primary demand driver is the marketplace itself: every purchase of a finished sound is denominated in SND, so the token is the medium of exchange of the platform.

Proposed SND allocation — 1,000,000 fixed supply

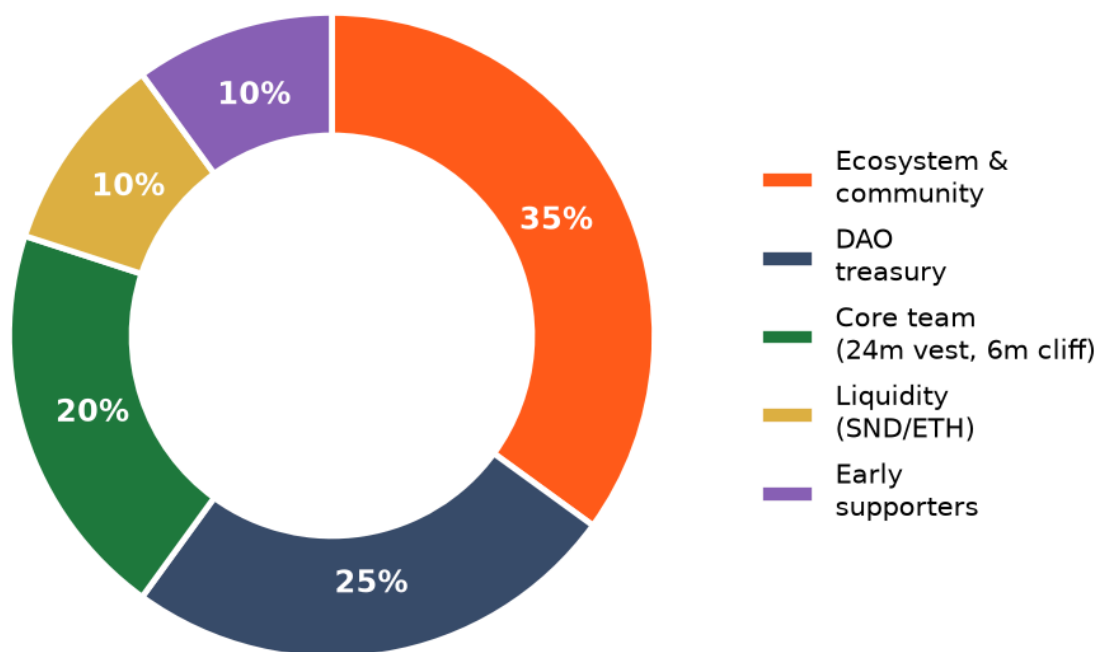


Figure 6 — Proposed allocation of the fixed 1,000,000 SND supply (subject to DAO ratification).

ALLOCATION	SHARE	PURPOSE
Ecosystem & community	35%	airdrops to producers, bounties, faucets
DAO treasury	25%	governed by SND holders via proposals
Core team	20%	24-month linear vesting, 6-month cliff
Liquidity	10%	SND/ETH pool on Base
Early supporters	10%	testnet contributors, early testers

The initial deployment vests the full supply with the deployer; distribution is **subject to a DAO vote** and should be finalised before mainnet.

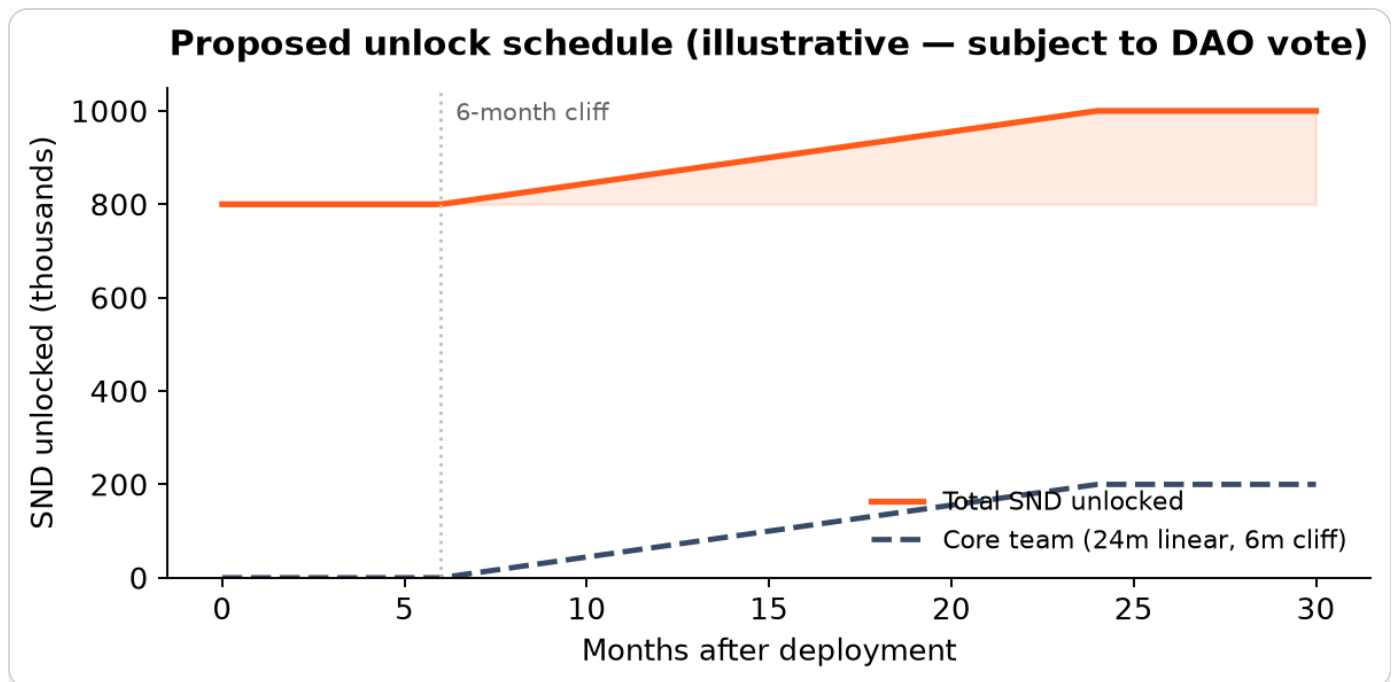


Figure 7 — Illustrative unlock schedule: 800,000 SND unlocked at deployment, team tranche linear over 24 months with a 6-month cliff.

The vesting schedule can be expressed as a piecewise function over months t since deployment, with cliff $c = 6$ and vesting end $v = 24$:

$$\text{unlocked}(t) = \begin{cases} 0, & t < c \\ \text{total} \cdot \frac{t - c}{v - c}, & c \leq t \leq v \\ \text{total}, & t > v \end{cases} \quad (6)$$

⚠ Nothing above is finalised.

The deployed testnet token has no pre-committed distribution; tokenomics are a DAO decision.

8. Governance — SoundHubGovernor

SND holders govern the platform through a standard OpenZeppelin Governor:

- Voting delay **1 day** (blocks), voting period **3 days**, quorum **4% of supply**.
- Execution through a **1-day timelock** — no rug-pull governance.

- Proposals control platform parameters: fees, premium tiers, treasury allocation, feature funding.

Governance lifecycle — SoundHubGovernor + 1-day timelock



Figure 8 — The governance lifecycle, from proposal to execution.

A proposal passes if the votes-for exceed the quorum threshold $\tau = 0.04 \cdot \text{supply}$ and votes-for exceed votes-against:

$$\text{passes} \iff \text{votes_for} > \text{votes_against} \wedge \text{votes_for} + \text{votes_against} \geq \tau \quad (7)$$

The timelock delay $d_{\text{lock}} = 1$ day between a successful vote and execution gives token holders a final window to react before any parameter change takes effect.

9. Distribution — SoundHub inside Ableton Live

A marketplace is only as good as its placement. The strongest distribution channel for SoundHub is the DAW itself — the point of maximum intent:

- Point of maximum intent.** The producer is already inside Ableton, already hearing what's missing. Buying a finished preset there is a 1-2 click action — no tab switch, no context break.
- Native panel, not a rewrite.** A Max for Live device (JS inside Live) reads the catalog straight from the chain via public RPC, reads the Live set BPM for

context-aware suggestions, and loads purchased assets into the project. Prototype shipped in `m4l/`.

- **Context-aware, not just a storefront.** BPM → genre matching today; key, tracks and devices (via the DAW engine) tomorrow — “bass presets for 128 BPM techno” instead of a flat list.
- **Invisible web3.** The user sees *Buy & Load*, never `approve/gas/RPC`. WalletConnect pairing or a backend relay keeps the blockchain under the hood.
- **One engine, three markets.** The same DAW parsers that verify assets for sale power the in-DAW recommendations and the smart diffs.

Without the integration SoundHub is a good niche marketplace; with it, it is part of the producer’s workflow — the difference between a website and a tool.

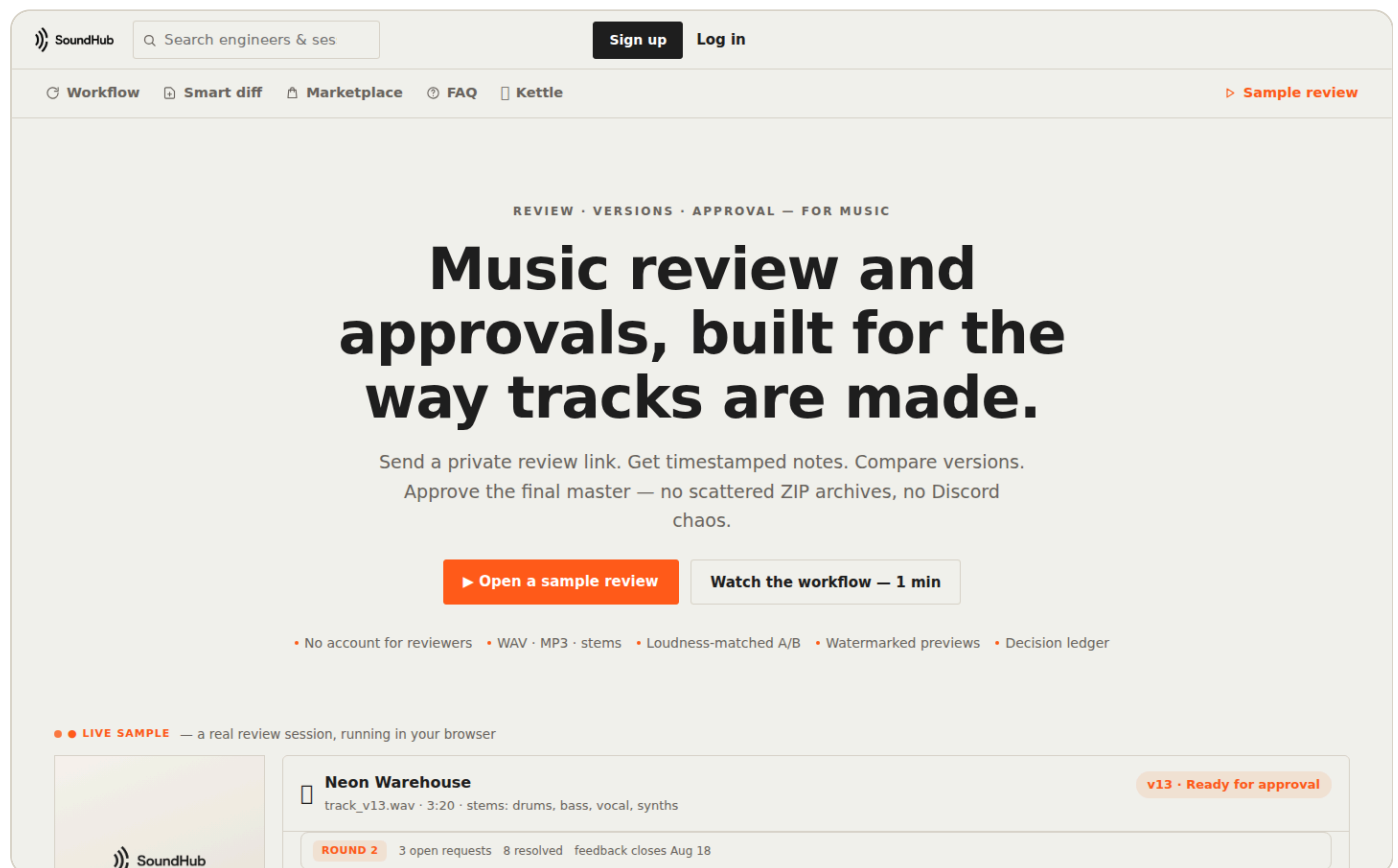


Figure 9 — The SoundHub landing page (bandcamp-inspired studio UI).

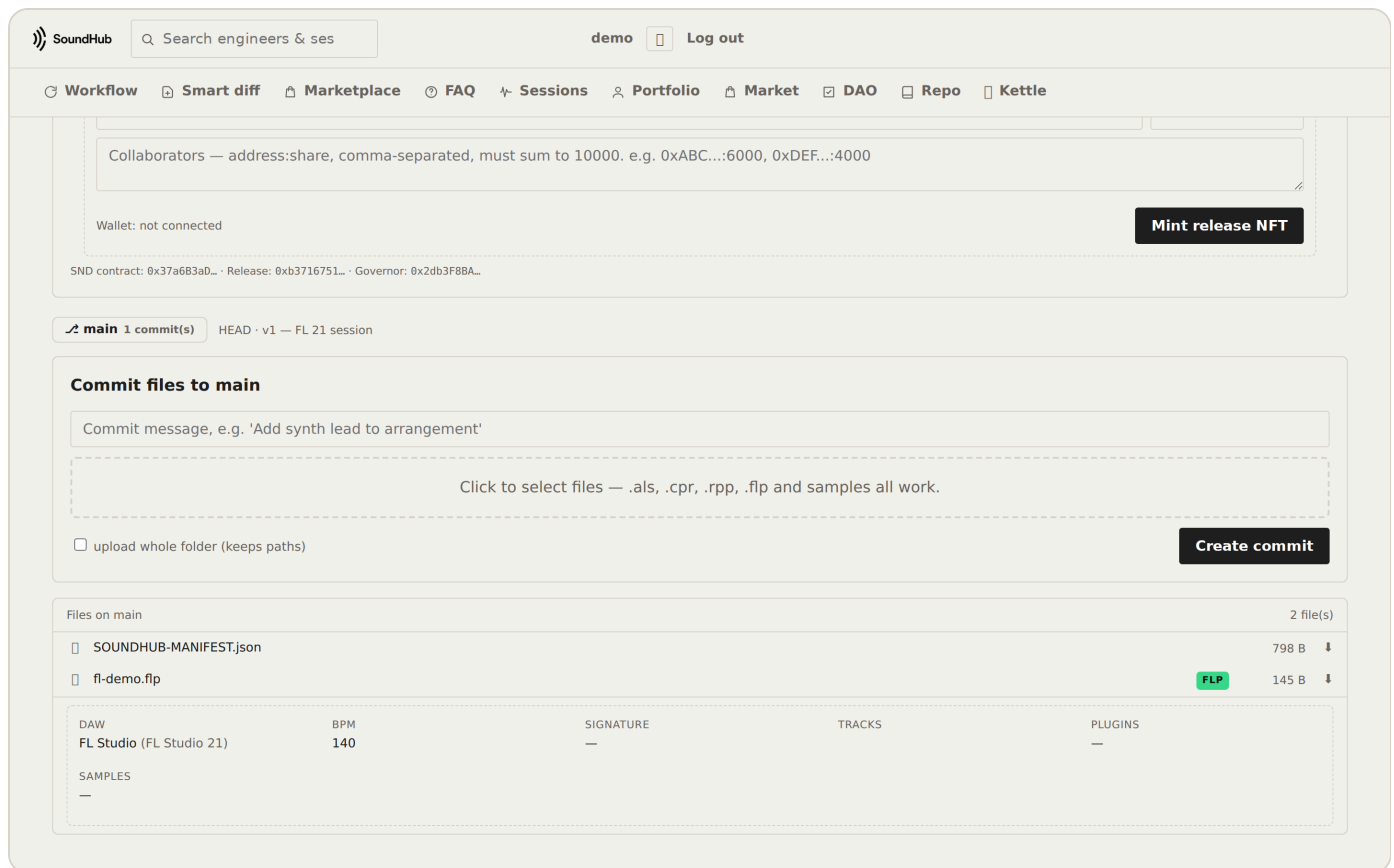


Figure 10 — A real FL Studio 21 project parsed in SoundHub (140 BPM, FL 21, three channels).

10. Revenue model

STREAM	MECHANISM
Marketplace fee	small % on each escrow purchase (configurable via DAO): funds development, curation, grants, ecosystem rewards
Sellers earn	the core creator income: finished sounds sold for SND, settled automatically
Premium tier	token-gated (hold/stake SND): deeper verification, larger listings, analytics
Release royalties	creator-set ERC-2981 royalty on Release NFT secondary sales (default 5%)
Tips	fans tip producers in SND or ETH, fully on-chain

The marketplace fee is the platform's primary recurring revenue. For a purchase of price p and fee rate f , the fee and the seller payout are:

$$\text{fee} = p \cdot f, \quad \text{seller_payout} = p \cdot (1 - f) \quad (8)$$

The fee rate f is a DAO-controlled parameter, deliberately kept small and transparent.

11. Security, risks & disclaimers

⚠ Unaudited contracts.

The smart contracts pass an extensive test suite (token, royalties, splits, full DAO lifecycle) but have **not** been through a professional security audit. Test on testnet first.

11.1 What is tested

- **SND token** — ERC-20 semantics, permit, votes, no-mint invariant, fixed supply.
- **SoundHubMarket** — escrow hold/release/refund, dispute window, arbiter paths.
- **SoundHubRelease** — royalties (ERC-2981), collaborator splits, order-independent treasury claims.
- **Governor** — full proposal → vote → timelock → execute lifecycle.
- **Backend** — 100+ tests covering the DAW parsers, diff engine, versioning and the `snd push` contract.

11.2 Risks

- **Narrative-driven market.** The value of SND is tied to platform adoption, not to any underlying utility guarantee.
- **Format complexity.** DAW files are reverse-engineered formats; parser coverage is best-effort and improves over time.
- **Concentration risk.** Governance uses a 4% quorum; a coordinated minority could in principle pass proposals during low turnout, mitigated by the timelock window.
- **Testnet data.** All on-chain contracts below are on **Base Sepolia**; addresses and behaviour are subject to change before mainnet.

📌 **Live today (honest).**

DAW engine (4 formats), smart diffs, repos & content-addressed versioning,
SND/Release/Market/Governor on Base Sepolia, escrow buy flow with faucet, wallet sign-in,
in-DAW M4L prototype.

12. Roadmap

PHASE	STATUS	SCOPE
1 • Buy flow	📌	SoundHubMarket live on Base Sepolia → list/buy/confirm/refund in the UI → SND faucet for testers
2 • In-DAW	📌 / 📌	SoundHub inside Ableton Live — M4L prototype shipped; next: recommendation service, asset delivery, full import
3 • Trust layer	📌	verification badges from the DAW engine, seller reputation, license enforcement, packs as composable assets
4 • Collaboration	📌 / 📌	branches (named pointers, per-branch diff) → merges (DAG), remix forks, audio preview, comments on tracks
5 • Mainnet	📌	Base mainnet deployment, SND distribution vote, DAO treasury

13. Contract addresses (Base Sepolia)

CONTRACT	ROLE	ADDRESS
SND (ERC-20)	platform token	0x37a6B3aD766ffb98673290A634490C8bF952DB2F
SoundHubRelease (NFT)	release NFTs, royalties, splits	0xb3716751572db83d22aeED95Be7da125A4d22446
SoundHubMarket (escrow)	escrow marketplace	0x396d6ad9D5EA19eE56318624b05bC6EEea2d1F5C
SoundHubFaucet (testnet)	testnet SND faucet	0x479fe2D308D118ef1723b5a34C8f8f37678cbba9
TimelockController	governance timelock	0x98F6a809ffa83cbe5f9bAFa7Cf762f2f24Cfa548
SoundHubGovernor (DAO)	governance	0x2db3F8BA478C445399bB8fbA921fC5e11Af202da

